Milestone 03

Answer Key

Environment

Update the YAML with your first and last name.

Load the packages.

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(haven)
library(gssr)
```

```
Package loaded. To attach the GSS data, type data(gss_all) at the console.
For the panel data and documentation, type e.g. data(gss_panel08_long) and data(gss_panel_doc
For help on a specific GSS variable, type ?varname at the console.
```

```r
library(gssrdoc)
library(summarytools)
```

```
Attaching package: 'summarytools'

The following object is masked from 'package:tibble':

    view
```

```r
library(Rmisc) # load the new package
```

```
Loading required package: lattice
Loading required package: plyr
------------------------------------------------------------------------------
You have loaded plyr after dplyr - this is likely to cause problems.
If you need functions from both plyr and dplyr, please load plyr first, then dplyr:
library(plyr); library(dplyr)
------------------------------------------------------------------------------

Attaching package: 'plyr'

The following objects are masked from 'package:dplyr':

    arrange, count, desc, failwith, id, mutate, rename, summarise,
    summarize

The following object is masked from 'package:purrr':

    compact
```

```r
library(gtsummary) # load the new package
```

```
Attaching package: 'gtsummary'

The following object is masked from 'package:plyr':

    mutate
```

Load the 2022 GSS data

```
# ADD YOUR CODE HERE TO LOAD THE GSS 2022 DATA
gss22 <- gss_get_yr(2022)
```

```
Fetching: https://gss.norc.org/documents/stata/2022_stata.zip
```

You'll be working with the following GSS variables:

- `id`: Respondent id
- `meovrwrk`: Men hurt family when focus on work too much
- `sex`: Respondent's gender
- `lifenow` R's rating of life overall now from 0-10
- `educ` Highest year of school completed

Code, Output, Meaning

Use R to complete the checkpoints below. Show your work (e.g., R code chunks) where appropriate. Add narrative (text outside code chunks) or comments (text inside the code chunks) throughout.

Show your df and variable changes as you go (e.g., `table()`, `head()`). Reference specific statistics (where appropriate) from your output to justify your answers. Explain what the values tell you about the data; interpret their meaning in relation to the question.

Checkpoint 01: Create a new dataframe

A. Create a df named `my_data` with `meovrwrk`, `sex`, `marital`, `lifenow`, and `educ`. Then, use `drop_na()` to remove rows with missing data.

```
my_data <- gss22 |>
  select(id, meovrwrk, sex, marital, lifenow, educ) |>
  drop_na()

head(my_data)
```

```
# A tibble: 6 x 6
     id meovrwrk                            sex        marital      lifenow   educ
  <dbl> <dbl+lbl>                           <dbl+lbl>  <dbl+lbl>    <dbl+lb> <dbl+lb>
1     4 2 [agree]                           2 [female] 5 [never ma~  8        16 [4 y~
2     9 3 [neither agree nor disagree]      2 [female] 5 [never ma~ 10 [10 ~ 12 [12t~
```

```
3     13 4 [disagree]                     2 [female] 1 [married]    8        12 [12t~
4     18 3 [neither agree nor disagree] 2 [female] 1 [married]    8        16 [4 y~
5     24 3 [neither agree nor disagree] 2 [female] 1 [married]    8        14 [2 y~
6     25 2 [agree]                      1 [male]   5 [never ma~   9        13 [1 y~
```

Checkpoint 02: Clean and create variables

A. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `hurt_num` that groups respondents as follows:

- 1 if `meovrwrk` is 'strongly agree' or 'agree'
- 0 if `meovrwrk` is equal to anything else

```
my_data <- my_data |>
  mutate(
    hurt_num = case_when(
      meovrwrk %in% 1:2 ~ 1,
      TRUE ~ 0
    ))

table(my_data$meovrwrk, my_data$hurt_num)
```

```
      0   1
  1   0 123
  2   0 535
  3 408   0
  4 308   0
  5  74   0
```

B. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `college` that groups respondents as follows:

- "No college degree" if `educ` is less than 16
- "College degree" if `educ` 16 or more

```
my_data <- my_data |>
  mutate(
    college = case_when(
      educ < 16 ~ "No college degree",
      educ >= 16 ~ "College degree"
  ))
```

```r
table(my_data$educ, my_data$college)
```

```
   College degree No college degree
  2              0                 1
  3              0                 1
  4              0                 1
  5              0                 3
  6              0                 5
  7              0                 3
  8              0                13
  9              0                10
  10             0                12
  11             0                36
  12             0               334
  13             0               108
  14             0               193
  15             0                90
  16           350                 0
  17            74                 0
  18           100                 0
  19            37                 0
  20            77                 0
```

C. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `edu_category` that groups respondents as follows:

- "High school or less" if `educ` is 12 or less
- "Some college" if `educ` is equal to or greater than 13 and equal to or less than 15
- "College or more" if `educ` is 16 or more

BONUS: Then, use `factor()` to put the categories into a meaningful order

```r
my_data <- my_data |>
  mutate(
    edu_category = case_when(
      educ <= 12 ~ "High school or less",
      educ >= 13 & educ <= 15 ~ "Some college",
      educ >= 16  ~ "College or more"
  ))

# Convert edu_category to an ordered factor
```

```r
my_data <- my_data |>
  mutate(
    edu_category = factor(
      edu_category,
      levels = c("High school or less", "Some college", "College or more")
      ))

table(my_data$educ, my_data$edu_category)
```

```
    High school or less Some college College or more
  2                   1            0               0
  3                   1            0               0
  4                   1            0               0
  5                   3            0               0
  6                   5            0               0
  7                   3            0               0
  8                  13            0               0
  9                  10            0               0
  10                 12            0               0
  11                 36            0               0
  12                334            0               0
  13                  0          108               0
  14                  0          193               0
  15                  0           90               0
  16                  0            0             350
  17                  0            0              74
  18                  0            0             100
  19                  0            0              37
  20                  0            0              77
```

D. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `marstat` that groups respondents as follows:

- "Married" if `marital` is 'married'
- "Single" if `marital` is 'never married'
- "Prev married" if `marital` is 'widowed', 'divorced', or 'separated'

BONUS: Then, use `factor()` to put the categories into a meaningful order

```r
my_data <- my_data |>
  mutate(
    marstat = case_when(
      marital == 1 ~ "Married",
      marital == 5 ~ "Single",
      marital %in% 2:4   ~ "Prev married"
  ))

# Convert edu_category to an ordered factor
my_data <- my_data |>
  mutate(
    marstat = factor(
      marstat,
      levels = c("Married", "Single", "Prev married")
      ))

table(my_data$marital, my_data$marstat)
```

```
    Married Single Prev married
  1     631      0            0
  2       0      0           22
  3       0      0          225
  4       0      0           39
  5       0    531            0
```

E. Use `zap_missing()`, `as_factor()`, and `droplevels()` to make `sex` and `meovrwkr` into clean factor variables.

```r
my_data$sex <- zap_missing(my_data$sex)
my_data$sex <- as_factor(my_data$sex)
my_data$sex <- droplevels(my_data$sex)

my_data$meovrwrk <- zap_missing(my_data$meovrwrk)
my_data$meovrwrk <- as_factor(my_data$meovrwrk)
my_data$meovrwrk <- droplevels(my_data$meovrwrk)
```

Checkpoint 03: Create and interpret confidence interval

A. Calculate the mean and 95% confidence interval for `lifenow`

```
my_data |>
  dplyr::summarise(
    n = n(),
    mean_lifenow = mean(lifenow),
    sd_lifenow = sd(lifenow),
    lowCI_lifenow = CI(lifenow, ci = 0.95)['lower'],
    hiCI_lifenow  = CI(lifenow, ci = 0.95)['upper']
    )
```

```
# A tibble: 1 x 5
      n mean_lifenow sd_lifenow lowCI_lifenow hiCI_lifenow
  <int>        <dbl>      <dbl>         <dbl>        <dbl>
1  1448         7.73       1.65          7.64         7.81
```

Interpret your confidence interval (e.g., say in plain language what the CI is telling us).

B. Calculate the mean and 99% confidence interval for `lifenow`

```
my_data |>
  dplyr::summarise(
    n = n(),
    mean_lifenow = mean(lifenow),
    sd_lifenow = sd(lifenow),
    lowCI_lifenow = CI(lifenow, ci = 0.99)['lower'],
    hiCI_lifenow  = CI(lifenow, ci = 0.99)['upper']
    )
```

```
# A tibble: 1 x 5
      n mean_lifenow sd_lifenow lowCI_lifenow hiCI_lifenow
  <int>        <dbl>      <dbl>         <dbl>        <dbl>
1  1448         7.73       1.65          7.61         7.84
```

How does the 95% confidence interval differ from the 99% confidence interval in terms of width and certainty?

Checkpoint 04: Create, compare and interpret confidence intervals

A. Calculate the mean and 95% confidence intervals for `lifenow` by `college`

```r
my_data |>
  group_by(college) |>
  dplyr::summarise(
    n = n(),
    mean = mean(lifenow),
    sd = sd(lifenow),
    lowCI = CI(lifenow, ci = 0.95)['lower'],
    hiCI  = CI(lifenow, ci = 0.95)['upper']
    )
```

```
# A tibble: 2 x 6
  college               n  mean    sd lowCI  hiCI
  <chr>             <int> <dbl> <dbl> <dbl> <dbl>
1 College degree      638  7.93  1.55  7.81  8.05
2 No college degree   810  7.57  1.71  7.45  7.69
```

Do the confidence intervals for these two groups overlap? What does that imply about the statistical significance of their difference?

B. Calculate the mean and 95% confidence intervals for `lifenow` by `edu_category`

```r
my_data |>
  group_by(edu_category) |>
  dplyr::summarise(
    n = n(),
    mean = mean(lifenow),
    sd = sd(lifenow),
    lowCI = CI(lifenow, ci = 0.95)['lower'],
    hiCI  = CI(lifenow, ci = 0.95)['upper']
    )
```

```
# A tibble: 3 x 6
  edu_category            n  mean    sd lowCI  hiCI
  <fct>               <int> <dbl> <dbl> <dbl> <dbl>
1 High school or less   419  7.56  1.61  7.41  7.72
2 Some college          391  7.57  1.81  7.39  7.75
3 College or more       638  7.93  1.55  7.81  8.05
```

Do the confidence intervals for these three groups overlap? What does that imply about the statistical significance of their difference?

Checkpoint 05: Create, compare and interpret confidence intervals

Calculate the mean and 95% confidence intervals for `hurt_num` by `marstat` & `sex` (e.g, married women, married men, single women, single, men, etc.)

```
my_data |>
  group_by(marstat, as_factor(sex)) |>
  dplyr::summarise(
    n = n(),
    mean = mean(hurt_num),
    sd = sd(hurt_num),
    lowCI = CI(hurt_num, ci = 0.95)['lower'],
    hiCI  = CI(hurt_num, ci = 0.95)['upper']
    )
```

```
`summarise()` has grouped output by 'marstat'. You can override using the
`.groups` argument.
```

```
# A tibble: 6 x 7
# Groups:   marstat [3]
  marstat      `as_factor(sex)`     n  mean    sd lowCI  hiCI
  <fct>        <fct>            <int> <dbl> <dbl> <dbl> <dbl>
1 Married      male               350 0.563 0.497 0.511 0.615
2 Married      female             281 0.349 0.477 0.293 0.405
3 Single       male               240 0.496 0.501 0.432 0.560
4 Single       female             291 0.419 0.494 0.362 0.476
5 Prev married male               112 0.518 0.502 0.424 0.612
6 Prev married female             174 0.368 0.484 0.295 0.440
```

Assess whether gender differences within each marital status group are likely to be statistically meaningful.

Checkpoint 06: Write a null and research hypothesis

Write a null and research hypothesis for the association between the variables `lifenow` and `college`

Checkpoint 07: Produce and interpret a two-sample t-tests

Use `t.test()` to produce a two-sided t.test of the variables `lifenow` and `college`

```
t.test(my_data$lifenow ~ my_data$college, alternative = "two.sided")
```

```

	Welch Two Sample t-test

data:  my_data$lifenow by my_data$college
t = 4.2025, df = 1418.3, p-value = 2.805e-05
alternative hypothesis: true difference in means between group College degree and group No co
95 percent confidence interval:
 0.1919569 0.5280400
sample estimates:
   mean in group College degree mean in group No college degree
                       7.927900                        7.567901
```

Explain what the test reveals about the statistical differences in average life satisfaction between the two groups. What do the results suggest about the null and research hypotheses?

Checkpoint 08: Create cross-tabs with chi-square tests

Use `tbl_cross()` and `add_p()` to create a cross-tab with a chi-square test for the variables `meovrwrk` and `college`

```
my_data |>
  tbl_cross(
    row = meovrwrk,
    col = college,
    percent = "column") |>
  add_p(source_note = TRUE)
```

Is there a statistically significant difference in agreement that men hurt the family when they focus on work too much by college degree?

	college		
	College degree	No college degree	Total
meovrwrk			
strongly agree	48 (7.5%)	75 (9.3%)	123 (8.5%)
agree	248 (39%)	287 (35%)	535 (37%)
neither agree nor disagree	178 (28%)	230 (28%)	408 (28%)
disagree	127 (20%)	181 (22%)	308 (21%)
strongly disagree	37 (5.8%)	37 (4.6%)	74 (5.1%)
Total	638 (100%)	810 (100%)	1,448 (100%)

Pearson's Chi-squared test, p=0.3

	sex		
	male	female	Total
meovrwrk			
strongly agree	76 (11%)	47 (6.3%)	123 (8.5%)
agree	298 (42%)	237 (32%)	535 (37%)
neither agree nor disagree	185 (26%)	223 (30%)	408 (28%)
disagree	124 (18%)	184 (25%)	308 (21%)
strongly disagree	19 (2.7%)	55 (7.4%)	74 (5.1%)
Total	702 (100%)	746 (100%)	1,448 (100%)

Pearson's Chi-squared test, p<0.001

Checkpoint 09: Calculate and interpret a chi-square test

Use `tbl_cross()` and `add_p()` to create a cross-tab with a chi-square test for the variables `meovrwrk` and `sex`

```
my_data |>
  tbl_cross(
    row = meovrwrk,
    col = sex,
    percent = "column") |>
  add_p(source_note = TRUE)
```

Is there a statistically significant gender difference in agreement that men hurt the family when they focus on work too much?

Checkpoint 10: Produce and interpret a Pearson's Correlation Coefficient (R)

Use `cor.test()` to test the correlation between `lifenow` & `educ`

```
cor.test(
  ~ lifenow + educ,
  data = my_data,
  method = "pearson",
  na.action = na.omit
)
```

```

	Pearson's product-moment correlation

data:  lifenow and educ
t = 4.9847, df = 1446, p-value = 6.956e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.07898701 0.18028011
sample estimates:
      cor
0.1299727
```

Describe the strength and direction of any correlation between education and life state.

IPUMS Data

To keep your Research Brief progress on track, you'll complete short exercises that correspond with the new course material using your own dataset as part of your milestones.

- Choose an interval-ratio (e.g, continuous or proportion) variable that you wish to consider as a **dependent variable** (DV).
- Create a summary table for this DV, including the total number of observations, mean, median, and mode.
- Pick or create a dichotomous variable (e.g., gender, two racial categorizes; two age categories; two countries; two years; etc.) in your dataset and create confidence intervals for each group for your DV. Compare and interpret the confidence intervals.
- Write a null and research hypothesis using these two variables.

- Conduct a two sample t-test (using `t.test()`) for your DV and dichotomous variable. Interpret your results. How do they relate to your hypotheses?
- Choose a second interval-ratio variable that you think may be associated with your DV.
- Use `cor.test()` to formally test for an association between the two continuous variables. Interpret your results.